

Bellabeat Fitness Tracker Case Study Report

Objective

Analyze the Bellabeat daily activity dataset using SQL, Tableau, and Python to identify user behavior patterns and provide business recommendations.

Introduction

This project analyzes the Bellabeat dailyActivity_merged.csv dataset to understand activity levels, calorie expenditure, and sedentary behavior.

Dataset

Dataset used: dailyActivity_merged.csv. Important columns include Id, ActivityDate, TotalSteps, TotalDistance, VeryActiveMinutes, FairlyActiveMinutes, LightlyActiveMinutes, SedentaryMinutes, and Calories.

Data Cleaning

Verified data types, converted ActivityDate to date format, checked missing values, checked duplicates, and validated numeric values.

SQL Analysis

Performed aggregations, GROUP BY, HAVING, CASE, subqueries, CTEs, window functions, running totals, and user performance analysis.

Tableau Dashboard

Built KPI cards, weekday charts, scatter plot (Steps vs Calories), Top 10 Active Users, and Top 10 Users by Calories.

Python Analysis

Used Pandas for EDA and Matplotlib for visualizations including weekday analysis, scatter plot, and top user charts.

Key Findings

1. Average daily steps are below 10,000.
2. More steps generally lead to higher calorie burn.
3. Sedentary minutes are high.
4. Activity differs by weekday.
5. A small group of users contributes the highest activity.

Recommendations

Encourage 10,000 steps/day, send inactivity reminders, introduce weekly challenges, personalize goals, and reward consistent activity.

Conclusion

The project demonstrates end-to-end data analysis using SQL, Tableau, and Python and provides actionable business insights.